



Thermal-fluid dynamic behavior and its effect on particle distribution and solidification characterization during the laser melting deposition of Ti-based composites

Leilei Wang¹ · Yifan Li¹ · Xingyun Yang¹ · Bowen Shi¹ · Zhuanni Gao¹ · Xiaohong Zhan¹

Received: 4 May 2023 / Accepted: 16 January 2024 / Published online: 3 February 2024

© The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2024

Abstract

Laser melting deposition (LMD) is an advanced repairing and remanufacturing technology that involves complex thermo-dynamic and kinetic characteristics in a molten pool. A thorough understanding of the thermal-fluid dynamic behavior of molten metal is crucial for investigating the distribution of reinforced particles and the characterization of solidification. This study proposes a three-dimensional transient thermal-fluid coupling model for describing thermal-fluid dynamic behavior and particle migration behavior within a molten pool during LMD. The simulated morphology of the deposition layer and temperature agree well with the experiments. The simulation results indicate that Marangoni convection induced by temperature gradient dominates the convection mode. The fluid velocity distribution along the scanning direction of the molten metal is showing M shape. Meanwhile, the dimension of molten pool increases slightly, and the Marangoni convection becomes more vigorous with higher heat input. The Marangoni motion tends to drive the TiC particle migration from the center to both lateral sides, resulting in a more uniformly distributed of TiC particle in deposition layer. Furthermore, the solidification parameters at solid–liquid interface are calculated, which used to assess the solidification microstructure in deposition layer.

Keywords Laser melting deposition · Numerical modelling · Thermal-fluid dynamic behavior · Particle distribution · Solidification characterization

1 Introduction

The widespread application of Ti-6Al-4 V alloys in engineering components with complex shapes is challenging due to low hardness and poor wear resistance [1, 2]. Under severe wear conditions, the service life and reliability of certain parts manufactured by Ti-6Al-4 V alloy (such as blades, impellers) are limited, which inhibits their full potential [3]. TiC has been widely recognized as an exceptionally suitable ceramic reinforcement material owing to its remarkable strength and stiffness, as well as its robust wetting capacity when interacting with Ti-6Al-4 V alloys. Hence, the utilization of TiC ceramics for enhancing the surface properties of titanium alloys has garnered significant attention. Laser

melting deposition (LMD) is an advanced repairing and remanufacturing technology extensively used in the surface modification, repairing, and manufacturing of aerospace components [4–7]. The laser violently interacts with the powder and matrix during LMD, which makes the powder and matrix melt and solidify rapidly [8]. The evolution and formation of the molten pool in LMD involves a complicated heat and mass transfer [9–11]. The existing volume of molten pool space is small (just a few millimeters), the existence time is very short (fraction of a second), and the whole process is carried out at a high temperature (up to 3000 K). Therefore, it is challenging to investigate the transient thermal-fluid dynamic behavior of molten pools during LMD by experiments [12, 13].

A thorough understanding of the thermal-fluid dynamic mechanisms is imperative for investigating the reinforced particle distribution and solidification characteristics of the molten pool. Computational fluid dynamics (CFD) simulation stands out as a significant methodology for exploring the thermal-fluid dynamic behavior [14–17]. The quantitative analysis of heat transfer and fluid flow during LMD

✉ Leilei Wang
meileiawang@gmail.com

¹ College of Materials Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China

by CFD methods has attracted increasing attention [18–20]. T. DebRoy et al. [21] established a transient thermal-fluid coupling model of laser melting deposition and noted that the major heat transfer mechanism in a molten pool was convection. Song et al. [22] established the finite-volume model to forecast the free surface of molten pool and solidification cladding dimensions. Vundru et al. [23] investigated the influence of spreading molten metal on deposition peculiarities during LMD. Their results showed that the calculated deposition geometry under different process parameters agreed well with the experiments. Li et al. [24] considered the effects of powder impact force, Marangoni force, and gravity on the fluid flow and obtained the evolution of the temperature field and velocity field in laser cladding.

The complex thermodynamic and kinetic characteristics in molten pool during LMD significantly affect the particle migration and solidification characterization, which strongly affects the service performance [25, 26]. Therefore, investigating the thermal-fluid transport in a molten pool and its effect on the particle distribution and solidification microstructure in laser melting deposition could effectively improve performance. Gu et al. [27] investigated the dynamic fluid flow peculiarities and particle migration behavior in the molten pool. Cui et al. [28] studied the influence of Marangoni convection on the fluid flow and migration characteristics of SiC nanoparticles in a molten pool, and their results illustrated that Marangoni convection is conducive to the embedding of SiC nanoparticles into the Al surface layer. Gao et al. [29] obtained the instantaneous temperature distribution and temperature curve of laser cladding by numerical simulation and investigated the dynamic solidification process by calculating the solidification parameters of the solid/liquid interface. However, the influence of Marangoni convection on heat transfer was not considered. Gan et al. [30] proposed an CFD model to investigate the heat and mass transfer during LMD of Co-based alloys. The crystal morphology and dimension of solidification structure were predicted by solidification parameters. They noted that Marangoni movement was the major mechanism for heat and mass transfer within molten pool. Song et al. [31] investigated the fluid behavior and solidification characteristics within the molten pool during the metal laser direct deposition process using a CFD simulation model. They elucidated the formation process of the deposition layer. The results of their study revealed a typical solidification microstructure, characterized by a transition from columnar to equiaxed morphology, which was attributed to temperature gradients and Marangoni convection. Geng et al. [31] investigated the thermal-fluid behavior and solidification characteristics in laser welding of aluminum alloy sheets. They established a quantitative relationship between thermal evolution and solidification microstructure and improve it by optimizing the process.

During the preparation process of a titanium-based composite coating reinforced with TiC particles, the high melting point of TiC as a ceramic phase can lead to the presence of unmelted TiC particles within the cladding layer. The distribution of these TiC particles can significantly influence the mechanical properties of the resulting cladding. The thermal flow behavior of the molten pool significantly affects the distribution of particles. Therefore, investigations of the convection mode and the thermal behavior within molten pool are necessary, especially the thermal-fluid transport and its effects on the solidification microstructure and TiC particle distribution. Solving the problem of unclear particle migration mechanism in molten pool during laser melting and deposition.

In this study, a three-dimensional transient thermal-fluid coupling model during the preparation of titanium-based composite coating reinforced with TiC ceramic particles using LMD, including solid, liquid, and gas phases, was proposed, and the accuracy was verified by experiments. The user-defined function (UDF) uses C language to define the powder impact, protective gas impact, powder addition, and thermophysical properties. The volume of fluid (VOF) means was adopted to track the free surface of molten pool. The thermal behavior, fluid convection, and shape of the molten pool during LMD for different process parameters were calculated. According to the calculated molten pool thermal behavior, the solidification parameters of the molten pool were calculated, which in turn assessed the solidification microstructure in the deposition layer. Based on the calculated velocity field, the distribution pattern of TiC ceramic reinforced particles within the molten pool, as well as their migration mechanism, has been elucidated.

2 Experimental section

In this study, the Ti-6Al-4 V alloy was cut into 90×90×8 mm cuboid specimens. The Ti-6Al-4 V alloy was polished with abrasive papers and cleaned with acetone. The commercial agglomerated and sintered Ti-6Al-4 V powder (45 ~ 105 μm) and TiC powder (60 ~ 150 μm) were mixed in a ball mill with a mixing time of 1.5 h, and the weight fraction of TiC powders in the mixture was 10%. The SEM micrographs of the TiC powders and mixed powders are depicted in Fig. 1.

The laser-deposited titanium composite coatings were fabricated by an LDM 8060 metal 3D printer equipped with five-axis CNC equipment, a continuous wave (CW) ytterbium fiber laser generator (Laserline, LFD 6000–6), a powder feeder (DPSF-2), a water cooling system, a hermetic chamber, and a numerical control system (SINUMERIK 828D), as shown in Fig. 2. The entire hermetic chamber was filled with argon during laser melting deposition to prevent

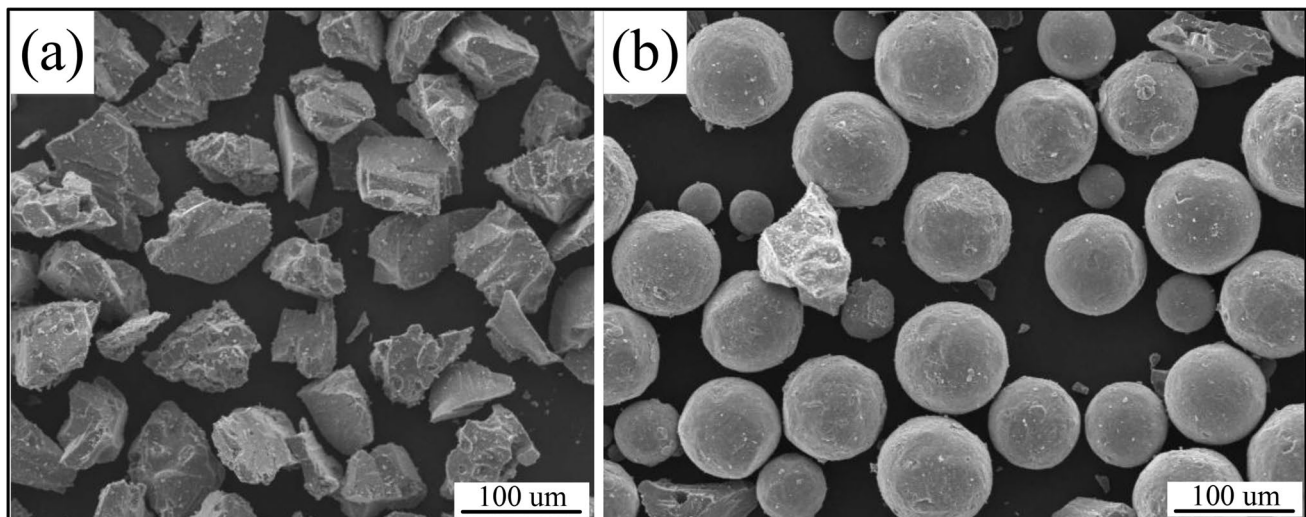


Fig. 1 SEM micrograph of different powders: **a** TiC **b** mixed powders with 10% TiC powder

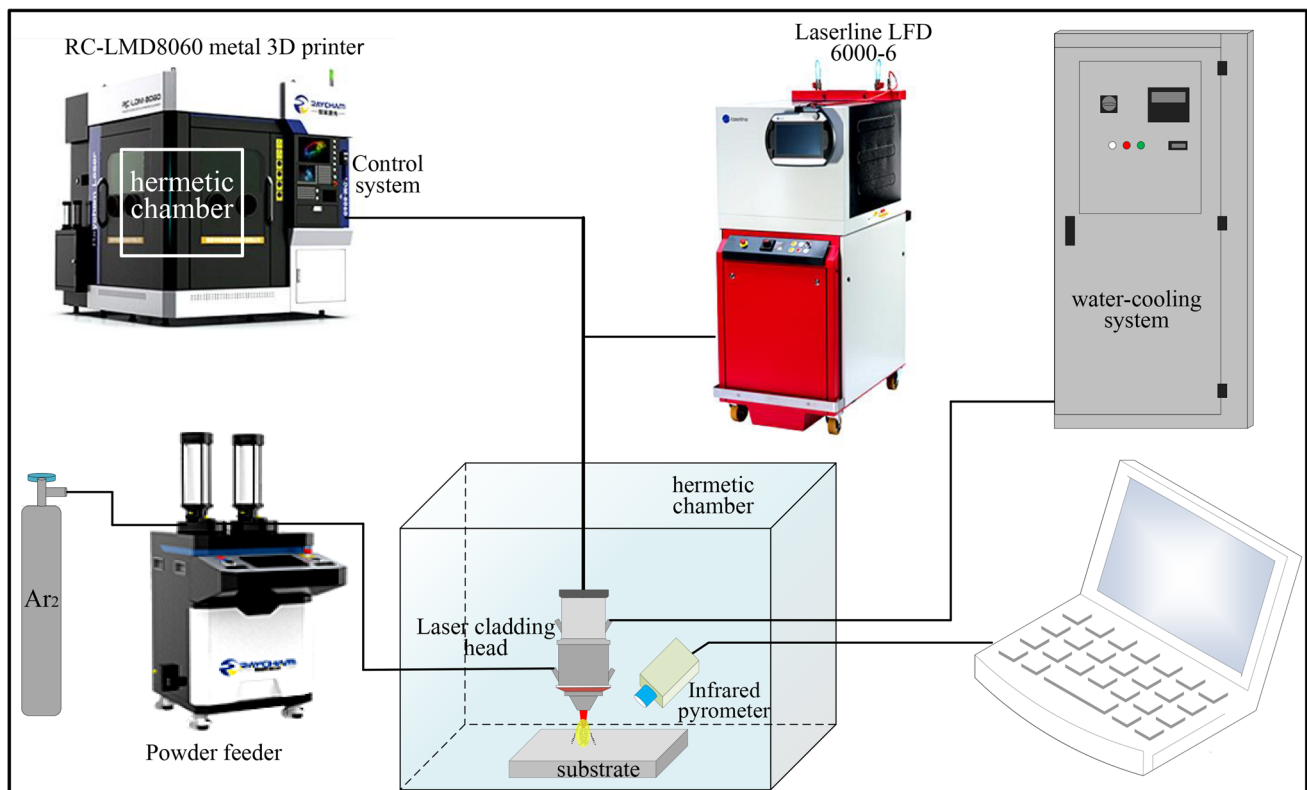


Fig. 2 Schematic diagram of the laser melting deposition

oxidation in the molten pool. According to the previous experimental results, laser powers of 1000 W, 1200 W, and 1400 W and a scan speed of 10 mm/s were selected for this study, as shown in Table 1. The laser deposited sample was cut in the direction perpendicular to the deposition path,

polished, and finally etched with the corrosive liquid consisting of 2% HF, 4% HNO₃, and 94% H₂O (volume %) for 10 s. The distribution of TiC particles and the microstructure of the deposited layer were analyzed using optical microscopy (MR-5000).

Table 1 The experimental parameters utilized in this study

Case	Laser power (W)	Scanning speed (mm/s)	Powder feeding rate (rad/min)	Beam diameter (mm)
1	1000	10	1.0	3.5
2	1200	10	1.0	3.5
3	1400	10	1.0	3.5

3 Numerical simulation modelling

3.1 Modelling assumptions

Owing to the complicated interactions between the heat source, powder, and metal, a number of assumptions ought to be made appropriately to simplify the CFD model. During LMD, the velocity and spatial distribution of powder particles are intricately linked to the process of gas-powder transportation. The impact of powder and carried gases can lead to surface fluctuations in the molten pool. Furthermore, the elevated temperatures within the molten pool may result in the combustion of certain elements, leading to a degradation in quality. The impact of powder and carried gas can cause fluctuations on the surface of the molten pool. In addition, the high temperature in the molten pool can also cause some elements to burn and cause quality loss. Since this article primarily addresses the flow behavior of the molten pool. The assumptions in this study are explicated as follows:

1. The distribution of powder particles is Gaussian after the interaction between laser and powder [18]. The powder has the same average velocity and temperature.
2. The carried gas has a stable and uniform velocity. This model merely considers the impact force motivated by carried gas of the powder delivery.
3. The laser power refers to the effective heat source absorbed by powder and metal. In this model, the absorptivity of the heat source is set at 35%.
4. The molten metal is assumed to be an incompressible Newtonian laminar flow fluid.
5. There is no chemical reaction within molten pool.
6. The metal is supposed to be isotropic and homogeneous.

3.2 Geometric model and material properties

The integrated calculation domain is $10 \times 6 \times 7$ mm, including gas and metal domains. Due to the characteristics of the computational domain far from molten pool having little impact on thermal-fluid dynamic behavior near the heat source, a full-scale computational domain does not need to be established [32]. Figure 3 depicts the particulars of the computational domain and mesh. In the finite volume

method, the accuracy and efficiency of calculation greatly depend on the size and number of elements. Therefore, the transition grid is adopted to guarantee computational accuracy and efficiency. The calculational domain is divided into 150,852 elements with a minimum of $0.1 \times 0.1 \times 0.1$ mm and a maximum of $0.2 \times 0.2 \times 0.2$ mm. The main thermophysical properties of the Ti-6Al-4 V and TiC are illustrated in Table 2, as calculated by professional software and correlative references [33–36]. Based on the thermophysical properties in Table 2 and mixing rules, the mass fraction averaged material properties are calculated by [37]:

$$\bar{x} = \alpha_1 x_1 + \alpha_2 x_2 \quad (1)$$

where α_1 and α_2 represent the mass fractions of Ti-6Al-4 V and TiC, respectively. $\bar{x}$ is the thermophysical properties of molten metal. x_1 and x_2 represent thermophysical properties of Ti-6Al-4 V and TiC, respectively.

3.3 Governing equations

3.3.1 Governing equations

The essence of the CFD calculation of LMD is energy and mass transfer within molten pool. Mass transfer, heat transfer, and fluid flow within molten pool are controlled by mass, energy, and momentum equations. They are expressed as following descriptions [18, 38]:

$$\frac{\partial \rho}{\partial t} + \nabla(\rho v) = S_{\text{mass}} \quad (2)$$

$$\frac{\partial H}{\partial t} + H \nabla v = \frac{1}{\rho} (\nabla \cdot k \nabla T) + \frac{h_s}{\rho} + S_{\text{energy}} \quad (3)$$

$$\frac{\partial v}{\partial t} + v \nabla v = u \nabla v + g - \frac{1}{\rho} \nabla p + S_{\text{mom}} \quad (4)$$

where ρ represents the density of metal, t represents the calculated time, and v is the velocity of the fluid. H is the enthalpy of the metal, k is the thermal conductivity, h_s is the increment enthalpy of the external filling material, p is the ambient pressure, g represents the acceleration due to the gravity, and u represents the viscosity of molten metal. S_{mass} represents the mass source term, S_{energy} represents the momentum source term, and S_{mom} represents the momentum source term.

3.3.2 Mass source terms

The mass source term is the synchronous powder addition source term. The powder interacts with the laser, absorbing heat to melt and enter molten pool.

Fig. 3 Schematic diagram and meshing of CFD simulation. **a** Schematic diagram of CFD simulation. **b** Mesh model of CFD simulation. **c** CFD simulation of initial phase distribution

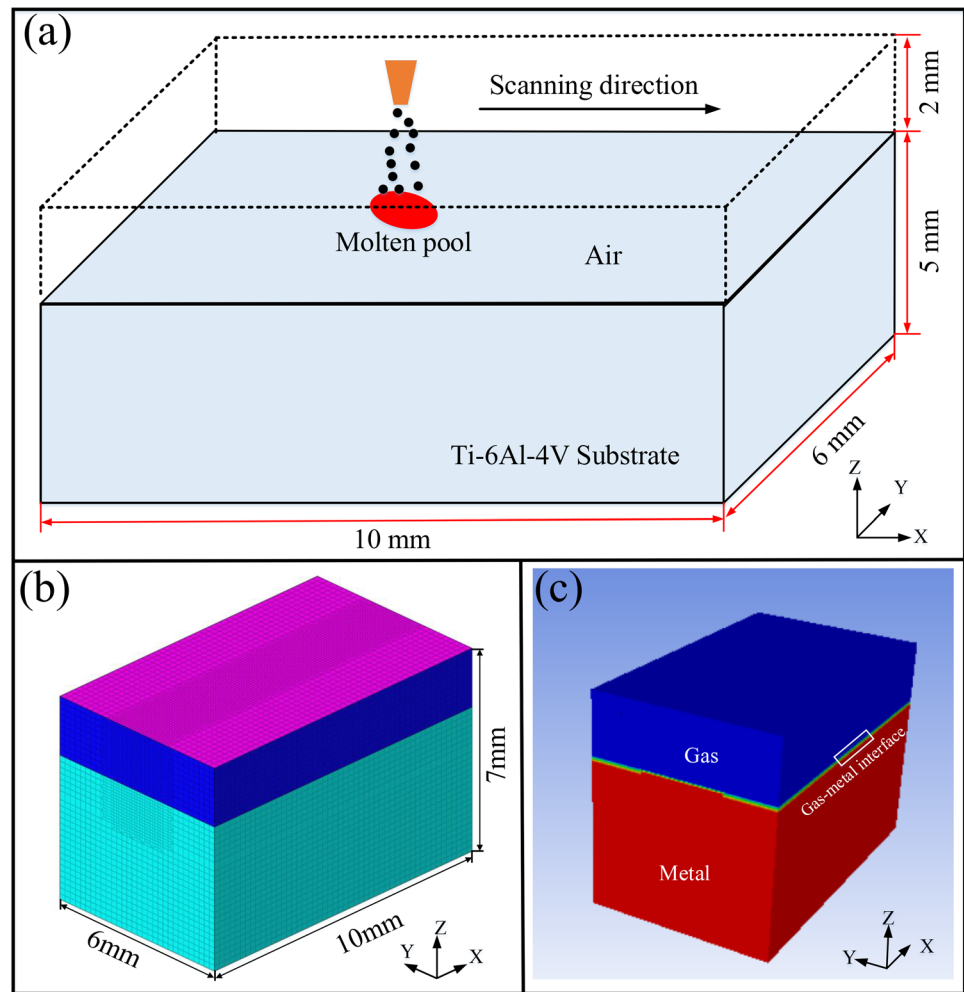


Table 2 Parameters used in the CFD model

	Symbol	Unit	Ti-6Al-4 V	TiC
Ambient temperature	T_{∞}	K	300	300
Solidus temperature	T_S	K	1878	3323
Liquidus temperature	T_L	K	1928	3353
Density	ρ	$\text{kg} \cdot \text{m}^{-3}$	4430	4930
Specific heat	L_m	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$	$492.5 + 0.025T - 4 \times 10^{-6}T^2$	796
Surface tension	γ_m	$\text{N} \cdot \text{m}^{-1}$	1.52	0.2
Thermal conductivity	K	$\text{W}^{-1} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	$1.57 + 1.6 \times 10^{-2}T - 1 \times 10^{-6}T^2$	$19.7 + 0.011 T$
Viscosity	μ_l	$\text{kg} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	0.004	0.05
Surface tension gradient	$\frac{\partial \gamma_m}{\partial T}$	$\text{N} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	-0.00026	0.0004
Stefan-Boltzmann constant	σ	$\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$	5.67×10^{-8}	
Latent heat	L	$\text{J} \cdot \text{kg}^{-1}$	2.86×10^5	
Radiation emissivity	ϵ	/	0.36	
Convective heat transfer coefficient	h	$\text{W} \cdot \text{m}^2 \cdot \text{K}$	80	

According to the assumptions, the spatial concentration distribution of the powder is similar to a Gaussian distribution, which is represented by the following function:

$$S_{\text{mass}} = \frac{\omega_1 m_s}{\pi r_b^2} \exp\left(-\frac{\omega_1 r^2}{r_b^2}\right) \quad (5)$$

where m_s is the mass flow of powder, and the value of this experiment is 8 g/min. ω_1 is the Gaussian distribution coefficient of the powder. r_b represents the radius of laser spot. r represents the distance from the laser spot center at this time, which can be described as follows:

$$r = \sqrt{(x - vt - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} \quad (6)$$

where x , y , and z represent the coordinate values of the cell center at time t , and x_0 , y_0 , and z_0 are the initial positions of LMD. The Gaussian distribution powder model is accumulated at each step in this model. As shown in Fig. 4, n_1 represents the height of the deposited powder at time t_1 , while n_2 represents the height of the deposited powder at time t_2 . The deposition layer is finally formed after accumulating multiple time steps.

3.3.3 Momentum source terms

The forces generated within molten pool mainly include the external force caused by surface tension σ , the buoyancy F_f caused by the temperature gradient, and the impact force F_g , and F_f caused by protective gas and powders on the deposition surface, respectively, as shown in Fig. 5.

Due to the different temperature distributions of liquid metal, its surface tension is also different, which will cause the convection of liquid metal, and the surface tension is as follows [39]:

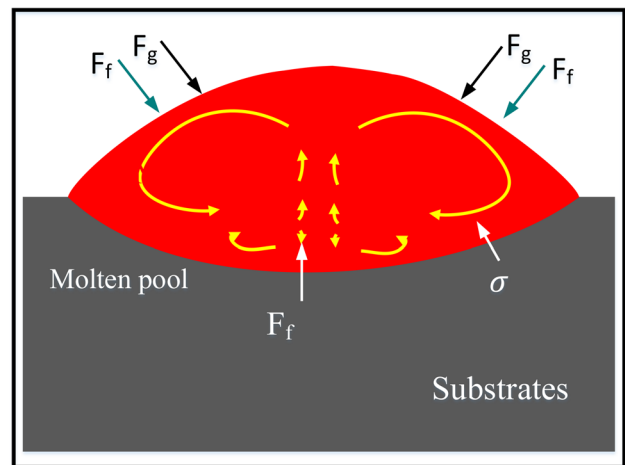


Fig. 5 The forces generated within molten pool

$$\sigma(T) = \sigma_0 - \sigma_T(T - T_1) \quad (7)$$

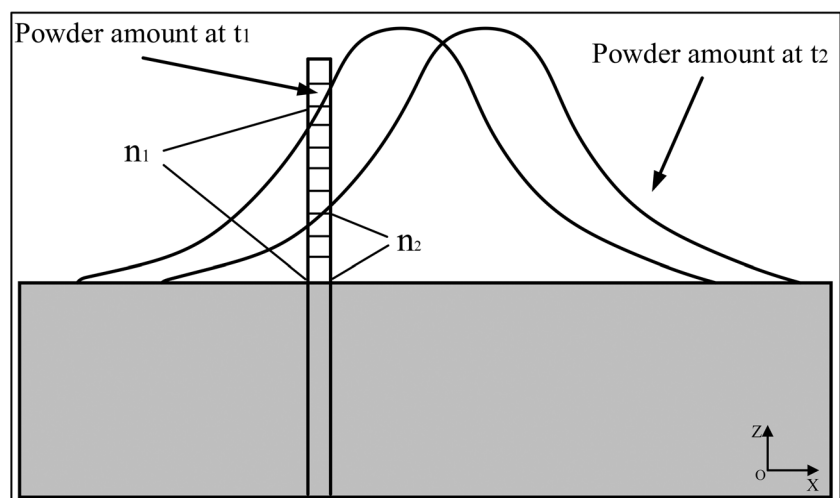
where $\sigma(T)$ represents surface tension between molten metal and gas, σ_0 represents the surface tension at the reference temperature, σ_T represents the coefficient of surface tension, and T represents the temperature of molten pool.

Furthermore, the thermal buoyancy of liquid metal is generally related to density, coefficient of thermal expansion and temperature variation. The following function represents the thermal buoyancy driving the natural convection of molten metal [40]:

$$F_f = \rho g \beta (T - T_1) \quad (8)$$

where T_1 is the liquidus temperature of the Ti-6Al-4 V titanium alloy and β represents the coefficient of thermal expansion. ρ represents the density of molten metal, and g represents the acceleration due to the gravity.

Fig. 4 Schematic diagram of powder stacking



It is assumed that the impact force generated by protective gas and powder is Gaussian distribution and acts on the surface of molten metal.

$$F_g = -2SP_g \exp\left(-\frac{r_1^2}{r_g^2}\right) \quad (9)$$

$$F_f = -2SP_f \exp\left(-\frac{r_2^2}{r_f^2}\right) \quad (10)$$

$$P_g = \rho_g v_g^2 \sin^2 \alpha \quad (11)$$

$$P_f = \rho_f v_f^2 \sin^2 \beta \quad (12)$$

where S is the unit area of surface of molten metal and P_g and P_f are the pressures caused by protective gas and powders, respectively. r_1 represents the distance from the molten metal surface to the center of the protective gas, r_2 represents the distance from the molten metal surface to the center of the powders, r_g represents the action radius of the protective gas, r_f represents the action radius of the powders, v_g represents the average velocity of the protective gas, v_f represents the average velocity of the powders, α represents the incident angle of the protective gas, and β represents the incident angle of the powders.

3.3.4 Energy source terms

As the only heat source in LMD, the molten pool moves with the laser beam. The heating conditions in different regions of the deposition layer are gradually differentiated with the laser beam moves, showing an apparent trailing phenomenon. The double-ellipsoidal heat source has many adjustable parameters, which is directly reflected in the obvious shape difference between the front and rear of the molten pool [41]. The heat flux of the front half is depicted as follows:

$$q_f(x, y, z) = \frac{6\sqrt{3f_f Q_a}}{a_f b c \pi \sqrt{\pi}} \exp\left(-3\frac{x^2}{a_f^2} - 3\frac{y^2}{b^2} - 3\frac{z^2}{c^2}\right) \quad (13)$$

Furthermore, the heat flux of the rear half is depicted as follows:

$$q_r(x, y, z) = \frac{6\sqrt{3f_r Q_a}}{a_r b c \pi \sqrt{\pi}} \exp\left(-3\frac{x^2}{a_r^2} - 3\frac{y^2}{b^2} - 3\frac{z^2}{c^2}\right) \quad (14)$$

where Q represents the laser power, α represents the absorption rate of laser, and a_f , a_r , b , and c are parameters of the heat flux. f_f and f_r are the distribution coefficients of the front and rear heat sources, respectively, which satisfy the limiting condition of $f_f + f_r = 2$, and f_f is 0.7, f_r is 1.3.

3.4 Initial conditions and boundary conditions

Figure 3a depicts the computational domain of numerical simulation which including gas and metal domains. The initial conditions of the whole domain are standard atmospheric pressure (101 kPa) and ambient temperature (300 K). The top surface of the gas domain is the pressure inlet, and the other surfaces are the pressure outlet. The substrate surfaces act as wall boundary including heat convection and heat radiation. The wall surface can be written as following equation [42]:

$$k \frac{\partial T}{\partial n} = -h_c(T - T_{\text{ref}}) - \epsilon \sigma (T^4 - T_{\text{ref}}^4) \quad (15)$$

where T and T_{ref} are the temperatures of the molten pool and ambient conditions, respectively. $\vec{n}$ is the normal vector along the wall boundary. σ represents the Stephan–Boltzmann constant, and ϵ represents the radiation emissivity of surface.

In addition, LMD involves both the melting and solidification of the metal. The enthalpy-porosity technique is employed to elucidate the phase transition behavior of the molten pool. The enthalpy-porosity technique. The enthalpy-porosity technique treats the solid–liquid two-phase region as a porous medium, with a porosity equal to the liquid volume fraction. This approach implicitly tracks the solid–liquid interface, thus simplifying the problem of metal phase transition.

4 Results and discussion

4.1 Simulation model validation

According to the thermal-fluid coupling model, CFD simulation during LMD of Ti-based composites was carried out, and the thermal behavior and flow field within molten pool were obtained. Figure 6 shows the transverse-section of deposition layers comparison between the experiments and simulation. The results show that the width and depth of the semi-deposition layer are in good agreement with the simulation. Furthermore, the molten pool temperature for different powers was measured. The results illustrate that the temperature increases with higher laser power and coincides with the temperature obtained by simulation. The simulations have a high degree of coincidence with the experiments.

4.2 Thermal-fluid dynamic behavior within the molten pool

Figure 7 illustrates the thermal-fluid dynamic behavior within the molten pool. Obviously, different regions in the molten pool have various convection modes. On the boundary of the molten pool, the molten metal flows from the leading and trailing edges to the bottom and rear of the molten

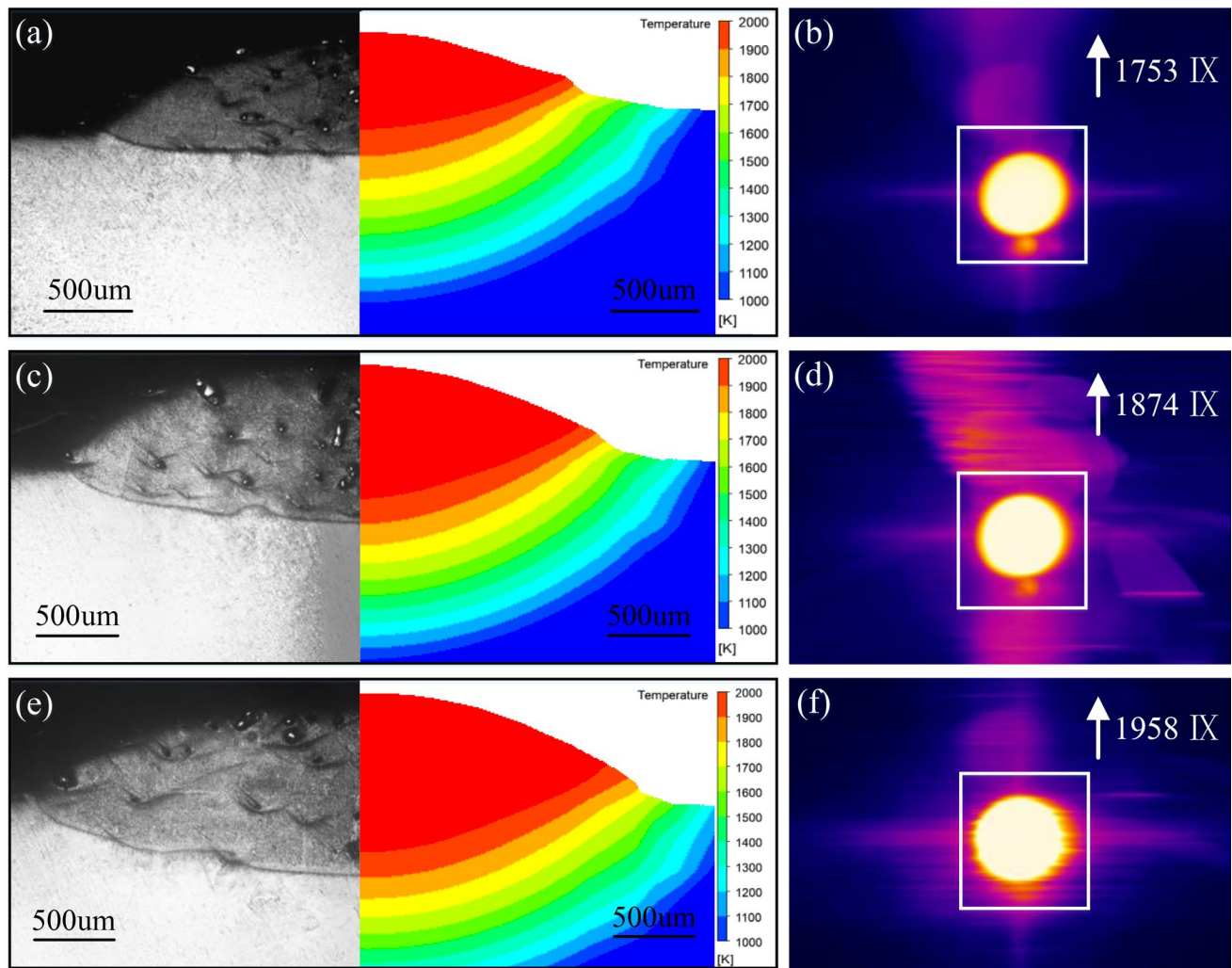


Fig. 6 The cross-sectional morphology and temperature comparison between simulation and experiment in different: **a, b** 1000 W; **c, d** 1200 W; **e, f** 1400 W

pool, as illustrated in Fig. 7a. Figure 7b illustrates the circular convection around the center of the molten pool parallel to the scanning direction. The molten metal at the leading end of the molten pool flows back to the center through the underside of molten pool, while some molten metal at the rear end of the molten pool expeditiously solidified and remains at the underside of the deposition layer, the others molten metal flows back to the center along the trailing end of the molten pool. The scale of the convection ring in front of the laser is much smaller than that behind the laser. Figure 7c shows that the molten metal moves radially outward around the center at the top surface. Based on the thermo-physical properties of the metal (as shown in Table 2), the surface tension gradient is consistently negative at both the upper and lower molten pool surfaces. Marangoni convection induced by the prominent spatial variation of the surface tension, which is driven by the temperature gradient along

the molten pool. This flow peculiarity in good agreement with previous study for two obvious circulation loops that dominate the molten pool caused by the Marangoni effect [39].

Figure 8 depicts the temperature distributions in the cross section and velocity distributions in the longitudinal section for case 2. Figure 8a and c show the location of monitoring points. In Fig. 8b, the temperature evolution at the monitoring points of the transverse section during the growth of the molten pool. With the displacement of the heat source, the temperature of the metal experiences a rapid increase, ultimately surpassing its melting point. After a period of time, the temperature of molten metal decreases slowly and starts to solidify, indicating that the temperature gradient is extremely sharp at the front of the laser beam and becomes flat away from the center of the heat source. A analogous variation tendency of

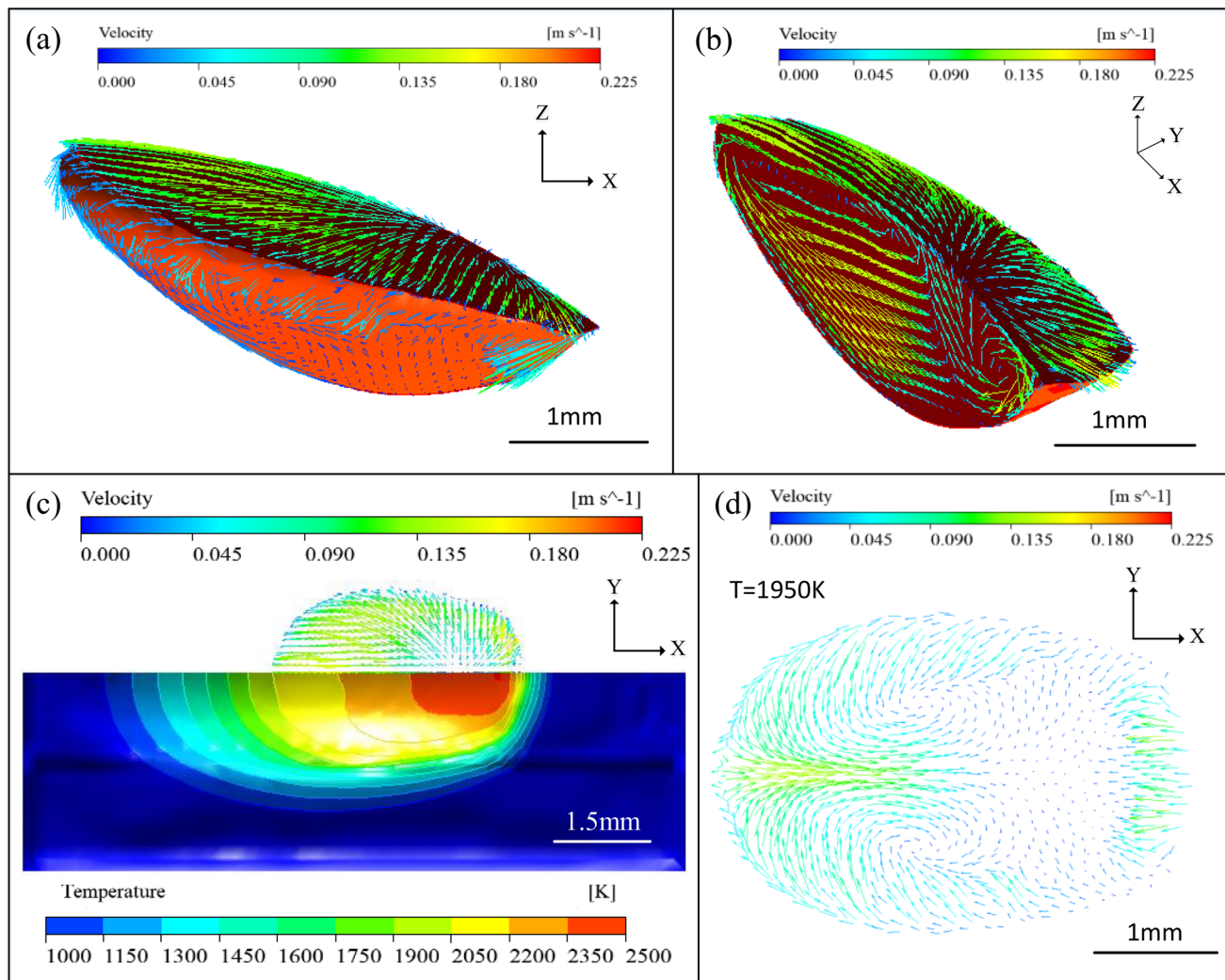


Fig. 7 The thermal-fluid dynamic behavior within the molten pool for case 2 ($P = 1200$ W, $v = 10$ mm/s): **a** convection model at the molten pool boundary; **b** convection model at the molten pool; **c** temperature

distribution and convection model on the top surface of the molten pool; **d** velocity field on the 1950 K isothermal surface

temperature is obtained at points 1–5. The highest temperature in the molten pool is close to 2400 K, which is higher than the melting temperature of titanium alloy (1928 K). Accordingly, it is concluded that the molten pool is formed rapidly and reaches a steady state. Figure 8d depicts the quantitative flow velocity distribution along the scanning direction in the stable molten pool. Based on the calculated results, the fluid flow within the molten pool exhibits two velocity peaks, forming a distinctive “M” shape pattern as it moves from the rear to the front end. The maximum flow velocity at the rear end of the molten pool is greater than that at the front end, which is in accord with the previous research [43]. Furthermore, the flow velocity in the molten pool decreases to the minimum due to the lower temperature gradient near the temperature peak.

4.3 Effects of process parameters

The laser power (P) is the most effective and accurately controlled parameter in laser melting deposition. As mentioned previously, three groups of experiments were conducted in which the laser power were 1000 W to 1200 W and 1400 W, respectively. The three-dimensional velocity field of the molten pool at a steady state for three laser powers is shown in Fig. 9. The steady-state widths of molten pool under three different laser powers are 2.8 mm, 3.3 mm, and 3.7 mm, respectively. Due to the consistent spot diameter, an escalation in laser power corresponds to an augmentation in heat input. Increased heat input results in enhanced catchment efficiency for capturing a greater amount of powder and expanding the dimensions of the

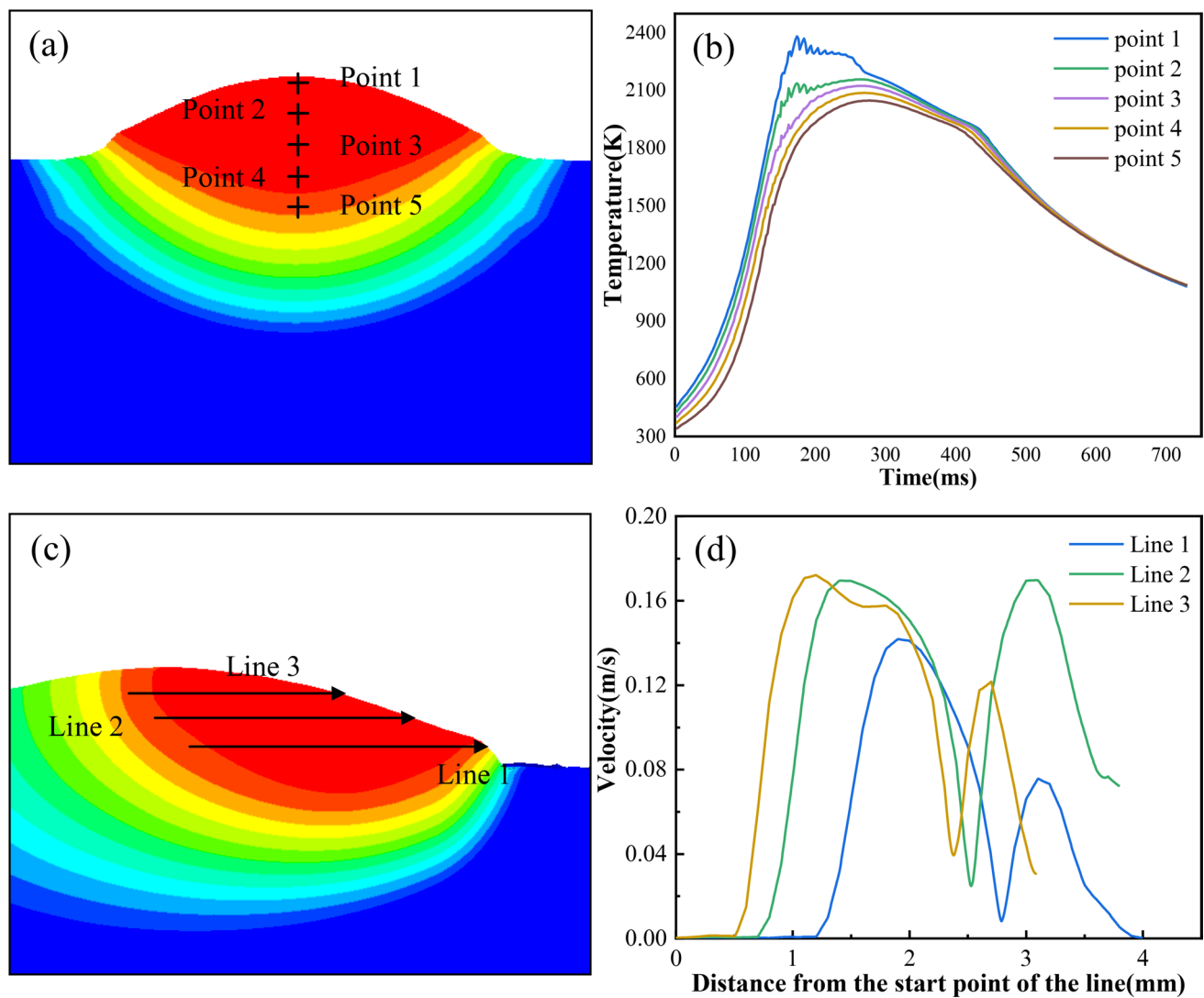


Fig. 8 Temperature distributions in the cross section and velocity distributions in the longitudinal section for case 2 ($P=1200$ W, $v=10$ mm/s): **a** the temperature field and monitoring points; **b** the

instantaneous temperature profile at monitoring points; **c** temperature field and monitoring lines; **d** velocity distributions at monitoring lines

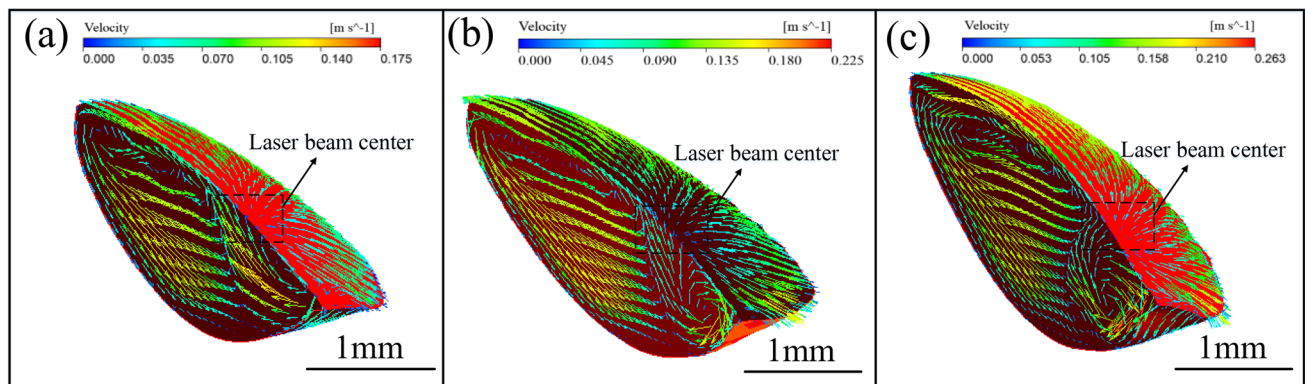


Fig. 9 Flow characteristics within molten pool with different process parameters: **a** $P=1000$ W, $v=10$ mm/s; **b** $P=1200$ W, $v=10$ mm/s; **c** $P=1400$ W, $v=10$ mm/s

molten pool. Therefore, the width and height of the deposition layer increase with the increase of laser power [44]. Furthermore, the lengths of the molten pool are 3.4 mm, 4.1 mm, and 4.7 mm, which can be used to investigate the solidification behavior. Since the scanning speed is 10 mm/s, it concludes that the solidifies time approaches 0.34 s, 0.41 s, and 0.47 s, respectively.

There are no substantial variations observed in the molten pool flow patterns when employing different sets of process parameters, as illustrated in Fig. 9. After the powder interacts with the laser, it undergoes melting and enters the molten pool. It flows from the center towards the edges before eventually entering the bottom of the molten pool. Due to the high thermal conductivity of the titanium alloy, the molten metal will experience partial solidification at the bottom of the melt. Meanwhile, the molten metal reflows to the surface due to the influence of the Marangoni movement [45]. Nevertheless, the velocity of fluid flow exhibits a slight increase with higher heat input. This phenomenon can be attributed to the amplified laser power, which in turn leads to a more pronounced temperature gradient and surface tension gradient. As a consequence, this gives rise to a more pronounced Marangoni convection effect [46]. Hence, the maximum flow velocity in the molten metal rises in tandem with the escalation of laser power.

4.4 Particle distribution and solidification microstructure

Figure 10 illustrates the cross-sectional morphology of the deposition layer fabricated with various laser powers. The results show good metallurgical bonding at the interface. It is obvious in Fig. 10a and b that TiC particles are mainly concentrated in the central region of the cross section. Due to the approximate density between TiC and the titanium matrix, gravity has little effect on the migration of TiC particles, which mostly relies on Marangoni convection [28]. This phenomenon can be attributed to three primary factors. Firstly, the Marangoni motion is relatively weak owing to the low laser power, and the particles can penetrate Marangoni convection. Secondly, there is no significant difference in temperature at the center of the cross-section. Therefore, the resultant force at the center of the cross section is approximately zero, limiting TiC particle migration in the molten pool. Thirdly, the cross-sectional view reveals the presence of dual convection rings, exhibiting nearly equal fluid velocities within the central region, albeit flowing in opposing directions. Consequently, the velocity magnitude at the cross section's center approximates zero [47]. However, as laser power increases, a notable shift occurs in the spatial distribution of TiC particles. The TiC particles move to both sides of

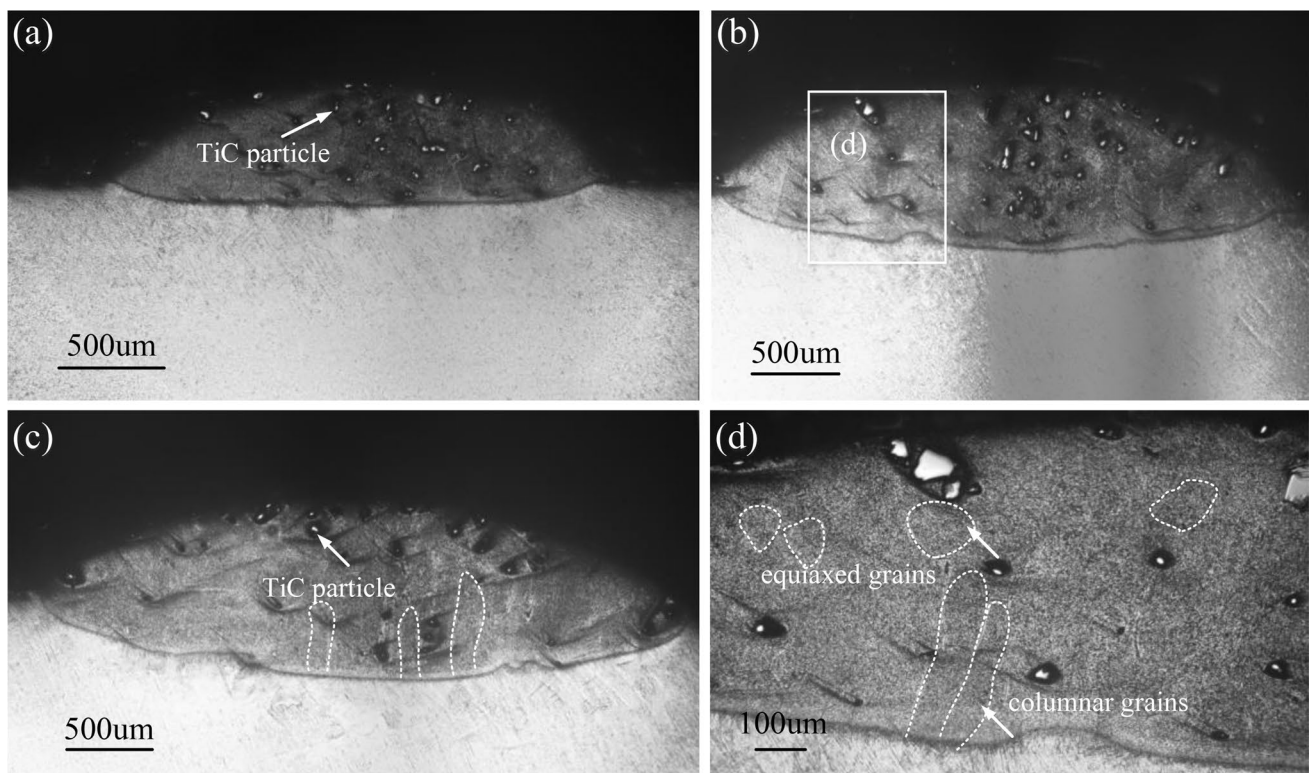


Fig. 10 The cross-sectional morphologies of coatings: **a** case 1; **b** case 2; **c** case 3; **d** micromorphology of the coating marked in **b**

molten pool and are uniformly distributed in the deposited layer, as illustrated in Fig. 10c.

The spatial distribution of TiC particles within the matrix is influenced by the interaction between the TiC particles and the molten pool. This interaction is governed by the thermal-fluid dynamic behavior occurring within the molten pool [48]. Figure 11 depicts the TiC particle distribution mechanisms within molten pool. The TiC particle get into molten pool with a certain speed and are mainly subjected to three kinds of external forces: gravity (F_g), buoyant force (F_f) and drag force (F_D), as shown in Fig. 11c [49].

According to classic multiphase flow theory, particles will be subject to fluid resistance due to the velocity difference between particles and fluid in the molten pool. The drag force can be described as follows [50]:

$$F_D = \frac{1}{6} \pi d_p^3 \rho_l \left(\frac{1}{\tau_p} \right) (u - v) \quad (16)$$

where F_D represents the drag force on the particle and τ_p represents the relaxation time, which is related to particle Reynolds number, Stokes drag force coefficient, fluid viscosity,

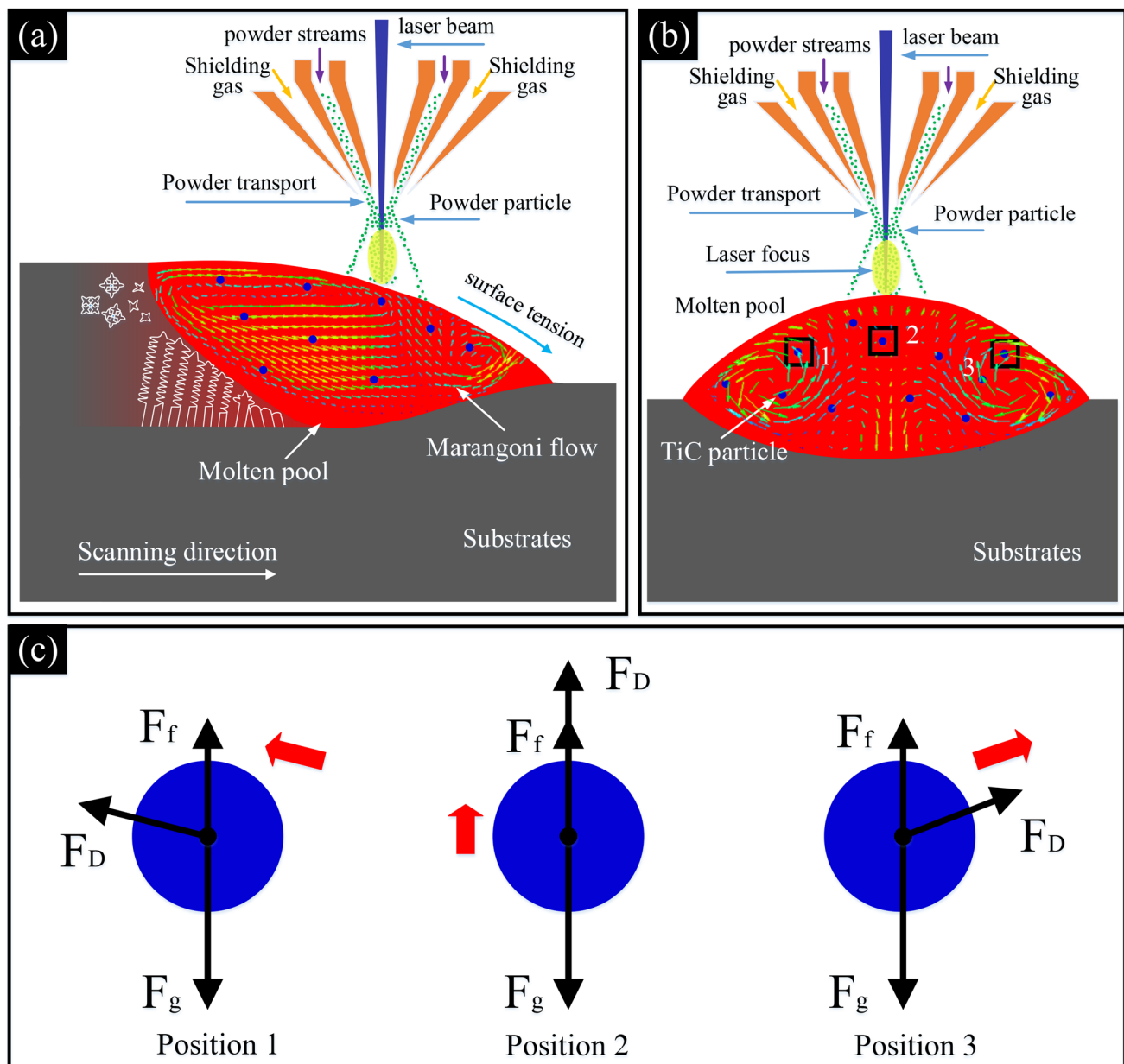


Fig. 11 Distribution mechanisms of the TiC particle within molten pool: **a** parallel to the scanning direction; **b** perpendicular to the scanning direction; **c** particle force analysis in different positions

etc. The shape and size of the particles are assumed to be the same. Thus, F_g and F_f are positively proportional. Consequently, the TiC particles will migrate with the melt under the influence of melt convection. Figure 11c shows a schematic diagram of the force acting on a particle in the molten pool. The particles at positions 1 and 3 are more likely to be dragged to edge due to the direction of Marangoni convection moving from the center to both sides of the molten pool. With the decrease of fluid flow and movement of solid–liquid interface, the TiC particle will eventually be engulfed, as shown in Fig. 11c. Owing to slow fluid flow and small drag force at position 2, the particles have difficulty entering the edge zone; thus, they tend to be trapped by the solidification interface.

Furthermore, the transformation of crystal growth morphology from bottom to top of the deposition layer was observed. The crystal growth morphology at the bottom is

columnar, and the top is equiaxed, as shown in Fig. 10d. Many studies have reported that the temperature gradient (G), growth rate (R), and their combination at the solid–liquid interface has significantly affect the solidification structure [32, 51, 52]. Figure 12 reveals the solidification parameters inside the rear end of the molten pool from top to bottom with different lines for case 2. The similar tendency of solidification parameters the surface to substrate along the molten pool for different lines. However, there are significant differences in solidification parameters from top to bottom within molten pool. The temperature gradient on the surface was low and even slightly reduced, which was attributed to the release of latent heat [53]. However, the temperature gradient increases rapidly with extension to middle of molten pool.

The higher temperature gradient and slower solidification rate at the lower portion of the deposition layer lead

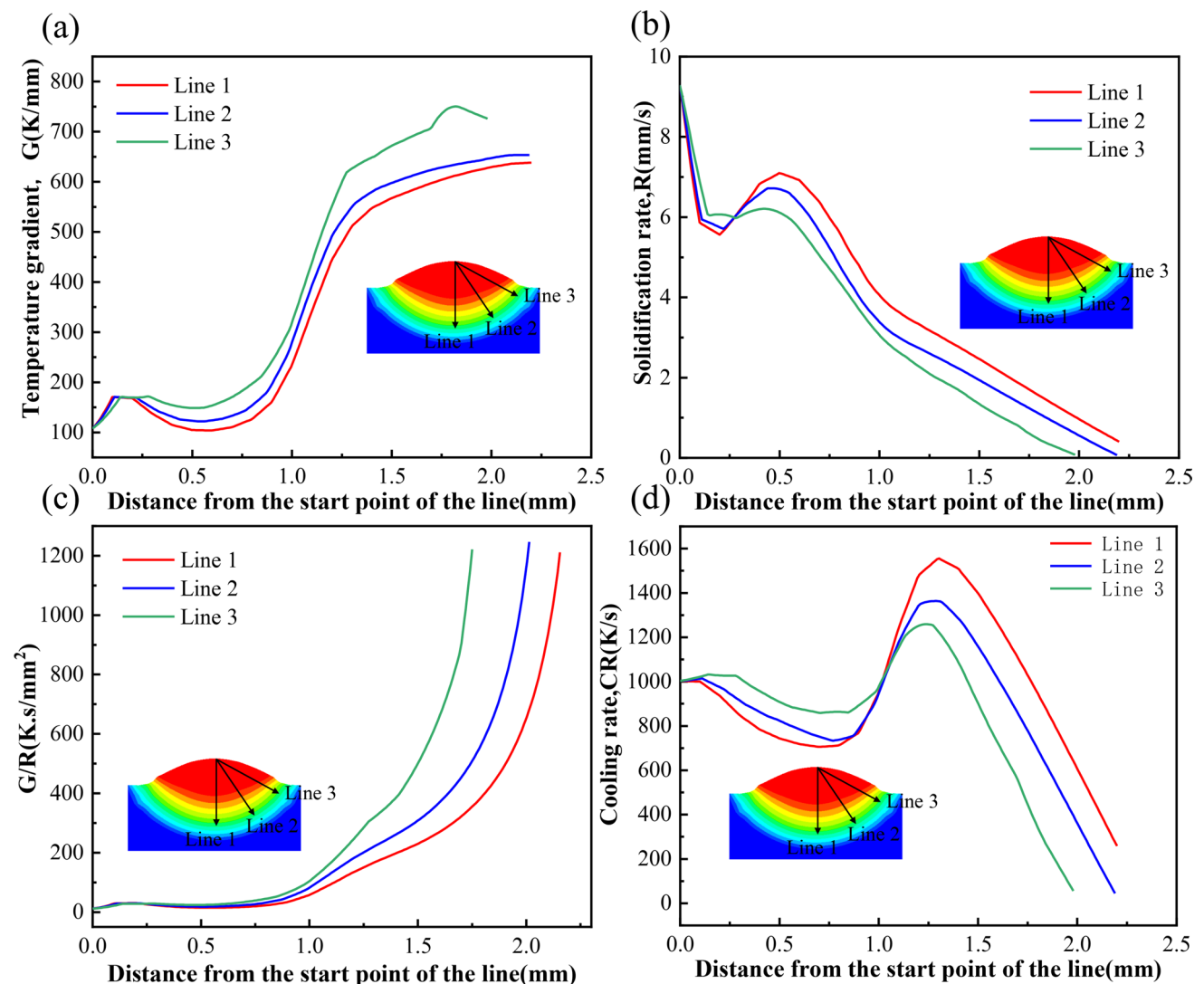


Fig. 12 The solidification parameters on different lines for case 2: **a** temperature gradient G ; **b** solidification rate R ; **c** G/R ; **d** GR

to an extended period for dendrite growth. Consequently, certain grains at the base preferentially grow onto the substrate grains, gradually inhibiting the growth of other grains and giving rise to the formation of columnar crystals along the temperature gradient. Conversely, the lower temperature gradient and accelerated solidification rate at the upper section of the deposition layer result in a briefer dendrite growth duration. As the solidification conditions near the front of the columnar crystal approach the critical threshold, a multitude of equiaxed crystals form atop the columnar crystals, hindering their growth progression. As a result, the grain morphology primarily exhibits an equiaxed crystals at the upper part of the deposition layer, while columnar crystals dominate at the lower regions.

5 Conclusion

We conducted CFD simulations to elucidate the thermal behavior and fluid convection within molten pool during LMD of Ti-based composites for different process parameters in this work. Furthermore, the thermal-fluid dynamic behavior and its effects on particle migration and solidification characterization were investigated. Subsequently, the solidification parameters were calculated according to the temperature distribution, and the solidification microstructure was assessed. The distribution pattern and the migration mechanism of TiC ceramic reinforced particles within the molten pool has been elucidated.

The major conclusions are as follows:

1. Considering the powder impact, protective gas impact, and powder addition in the molten pool, using the VOF method to track the free liquid surface of the molten pool, establishing a three-dimensional transient thermal-fluid coupling model for LMD. The thermal-fluid coupling model was validated and consistent with the experiments.
2. The temperature distribution and fluid convection of the molten pool during LMD for different process parameters were calculated. The Marangoni convection induced by the gradient of temperature and surface tension dominates the convection mode, and the fluid velocity distribution along the scanning direction showing M shape. The dimension of molten pool increased slightly, and Marangoni convection became more vigorous with increasing laser power.
3. The calculated solidification parameters were used to appraise the grain morphology. Due to the high temperature gradient and low solidification rate at the bottom against the low temperature gradient and high solidification rate at the top, the morphology of grains at middle and lower part of deposition layer are columnar, while the morphology of grains at top of deposition layer are equiaxed.
4. There exists an intense interaction between melt flow and TiC particles. The Marangoni motion tends to drive the TiC particle migration from the center to both lateral sides. The distribution of TiC particles is more uniform in the deposition layer with increased laser power.

Author contribution Leilei Wang: investigation, data curation, writing—original draft. Yifan Li: methodology, writing—review and editing. Xingyun Yang: methodology, writing—review and editing. Bowen Shi: data curation, investigation. Zhuanni Gao: data curation, investigation. Xiaohong Zhan: supervision, writing—review and editing.

Funding Financial support is provided by the Civil aerospace technology advance research project (D020304).

Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Kistler NA, Corbin DJ, Nassar AR, Reutzel EW, Beese AM (2019) Effect of processing conditions on the microstructure, porosity, and mechanical properties of Ti-6Al-4V repair fabricated by directed energy deposition. *J Mater Process Tech* 264:172–181
2. Zhao K, Zhang GH, Ma GY, Shen C, Wu DJ (2020) Microstructure and mechanical properties of titanium alloy/zirconia functionally graded materials prepared by laser additive manufacturing. *J Manuf Process* 56:616–622
3. Weng F, Chen CZ, Yu HJ (2014) Research status of laser cladding on titanium and its alloys: a review. *Mater Design* 58:412–425
4. Schubert E, Seefeld T, Rinn A, Sepold G (1999) Laser beam cladding: a flexible tool for local surface treatment and repair. *J Therm Spray Technol* 8:590–596
5. Qi CQ, Zhan XH, Gao QY, Liu LJ, Son YZ, Li YP (2019) The influence of the pre-placed powder layers on the morphology, microscopic characteristics and microhardness of Ti-6Al-4V/WC MMC coatings during laser cladding. *Opt Laser Technol* 119:105572
6. Wang YP, Ma SY, Yang XS, Zhou YZ, Liu X, Li JF, Zhang JJ, Li C, Wang XY, Le GM, Zhang Y (2020) Microstructure and strengthening mechanisms of 90W-7Ni-3Fe alloys prepared using laser melting deposition. *J Alloy Compd* 838:155545
7. Gao ZN, Bu HC, Feng Y, Lyu FY, Zhan XH (2021) Strengthening mechanism of Y2O3 nanoparticles on microstructure and mechanical properties of the laser additive manufacturing joint for large thickness TC4 titanium alloy. *J Manuf Process* 71:37–55
8. Aggarwal A, Patel S, Vinod AR, Kumar A (2021) An integrated Eulerian-Lagrangian-Eulerian investigation of coaxial gas-powder flow and intensified particle-melt interaction in directed energy deposition process. *Int J Therm Sci* 166:106963
9. Gao ZN, Wang LL, Lyu FY, Zhang YM, Liu T, Zhan XH (2022) Temperature variation and mass transport simulations of invar

- alloy during continuous-wave laser melting deposition. *Opt Laser Technol* 152:108163
10. Bayat M, Nadimpalli VK, Biondani FG, Jafarzadeh S, Thorborg J, Tiedje NS, Bissacco G, Pedersen DB, Hattel JH (2021) On the role of the powder stream on the heat and fluid flow conditions during Directed Energy Deposition of maraging steel-Multiphysics modeling and experimental validation. *Addit Manuf* 43:102021
11. Meng QB, Zhou XL, Li JH, Cui ZQ, Wang YD, Zhang HL, Li Z, Qiu CL (2020) High-throughput laser fabrication of Ti-6Al-4V alloy: Part I. Numerical investigation of dynamic behavior in molten Pool. *J Manuf Process* 59:509–522
12. Raghavan A, Wei HL, Palmer TA, DebRoy T (2013) Heat transfer and fluid flow in additive manufacturing. *J Laser Appl* 25:052006
13. Gao SM, Feng YC, Wang JJ, Qin M, Bodunde OP, Liao WH, Guo P (2021) Molten pool characteristics of a nickel-titanium shape memory alloy for directed energy deposition. *Opt Laser Technol* 142:107215
14. Zhang YM, Lim CWJ, Tang C, Li B (2021) Numerical investigation on heat transfer of melt pool and clad generation in directed energy deposition of stainless steel. *Int J Therm Sci* 165:106954
15. Du J, Wu YX, Wang DQ, Jiang MB, Wei ZY (2022) Droplet impact-induced molten pool dynamics and particle migration patterns during TIG-assisted droplet deposition manufacturing of SiC particle-reinforced Al-Si alloy. *Case Stud Therm Eng* 33:101962
16. Xu B, Jiang P, Wang Y, Zhao J, Geng S (2022) Multi-physics simulation of wobbling laser melting injection of aluminum alloy with SiC particles: SiC particles gradient distribution in fusion zone. *Int J Heat Mass Tran* 82:121960
17. Lyu FY, Wang LL, Feng Y, Gao ZN, Zhan XH, Wang XM (2021) Thermal behavior and microstructure evolution mechanism of Ti6Al4V 80 mm thick plates jointed by laser melting deposition. *J Manuf Process* 71:12–26
18. Zhao JX, Wang G, Wang XY, Luo S, Wang LP, Rong YM (2020) Multicomponent multiphase modeling of dissimilar laser cladding process with high-speed steel on medium carbon steel. *Int J Heat Mass Tran* 148:118990
19. Yao XX, Ge P, Li JY, Wang YF, Li T, Liu WW, Zhang Z (2020) Controlling the solidification process parameters of direct energy deposition additive manufacturing considering laser and powder properties. *Comp Mater Sci* 182:109788
20. Shi XY, Gu DD, Li YZ, Dai DH, Ge Q, Sun YX, Chen HY (2021) Thermal behavior and fluid dynamics within molten pool during laser inside additive manufacturing of 316L stainless steel coating on inner surface of steel tube. *Opt Laser Technol* 138:106917
21. Manvatkar V, De A, DebRoy T (2015) Spatial variation of melt pool geometry, peak temperature and solidification parameters during laser assisted additive manufacturing process. *Mater Sci Tech-Lond* 31:924–930
22. Song J, Chew YX, Bi GJ, Yao XL, Zhang BC, Bai JM, Moon SK (2018) Numerical and experimental study of laser aided additive manufacturing for melt-pool profile and grain orientation analysis. *Mater Design* 137:286–297
23. Vundru C, Singh R, Yan WY, Karagadde S (2021) Effect of spreading of the melt pool on the deposition characteristics in laser directed energy deposition. *Procedia Manuf* 53:407–416
24. Li C, Yu ZB, Gao JX, Zhao JY, Han X (2019) Numerical simulation and experimental study of cladding Fe60 on an ASTM 1045 substrate by laser cladding. *Surf Coat Tech* 357:965–977
25. Lee YS, Zhang W (2016) Modeling of heat transfer, fluid flow and solidification microstructure of nickel-base superalloy fabricated by laser powder bed fusion. *Addit Manuf* 12:178–188
26. Song BX, Yu TB, Jiang XY, Xi WC, Lin XL, Ma ZL, Wang Z (2022) Development of the molten pool and solidification characterization in single bead multilayer direct energy deposition. *Addit Manuf* 49:102479
27. Yuan PP, Gu DD, Dai DH (2015) Particulate migration behavior and its mechanism during selective laser melting of TiC reinforced Al matrix nanocomposites. *Mater Design* 82:46–55
28. Cui CY, Li XD, Fang C, Zhang WL, Ruan ZW, Cui XG, Lu JZ, Xia CD, Lu YF (2018) Effects of Marangoni convection on the embedding dynamic behavior of SiC nano-particles into the Al molten pool during laser micro-melting. *Mater Design* 143:256–267
29. Gao WY, Zhao SS, Wang YB, Zhang ZY, Liu FL, Lin XC (2016) Numerical simulation of thermal field and Fe-based coating doped Ti. *Int J Heat Mass Tran* 92:83–90
30. Gan ZT, Yu G, He XL, Li SX (2017) Numerical simulation of thermal behavior and multicomponent mass transfer in direct laser deposition of Co-base alloy on steel. *Int J Heat Mass Tran* 104:28–38
31. Geng SN, Jiang P, Shao XY, Guo LY, Gao XS (2020) Heat transfer and fluid flow and their effects on the solidification microstructure in full-penetration laser welding of aluminum sheet. *J Mater Sci Technol* 46:50–63
32. Song BX, Yu TB, Jiang XY, Xi WC, Lin XL (2021) Development mechanism and solidification morphology of molten pool generated by laser cladding. *Int J Therm Sci* 159:106579
33. Wei H, Mukherjee T, Zhang W, Zuback J, Knapp G, De A, DebRoy T (2021) Mechanistic models for additive manufacturing of metallic components. *Prog Mater Sci* 116:100703
34. Liu FQ, Wei L, Shi SQ, Wei HL (2020) On the varieties of build features during multi-layer laser directed energy deposition. *Addit Manuf* 36:101491
35. Wen SY, Shin YC (2011) Modeling of transport phenomena in direct laser deposition of metal matrix composite. *Int J Heat Mass Tran* 54:5319–5326
36. Li SC, Liu DF, Mi HB, Deng ZX, Liu J, Chen T (2022) Numerical simulation on evolution process of molten pool and solidification characteristics of melt track in selective laser melting of ceramic powder. *Ceram Int* 48:18302–18315
37. Dai DH, Gu DD, Ge Q, Li YZ, Shi XY, Sun YX, Li SH (2020) Mesoscopic study of thermal behavior, fluid dynamics and surface morphology during selective laser melting of Ti-based composites. *Comp Mater Sci* 177:109598
38. Voller VR, Prakash C (1987) A fixed grid numerical modelling methodology for convection-diffusion mushy region phase-change problems. *Int J Heat Mass Tran* 30:1709–1719
39. Gan ZT, Yu G, He XL, Li SX (2017) Surface-active element transport and its effect on liquid metal flow in laser-assisted additive manufacturing. *Int Commun Heat Mass* 86:206–214
40. Chen S, Zhao YQ, Tian SH, Gu YZ, Zhan XH (2020) Study on keyhole coupling and melt flow dynamic behaviors simulation of 2219 aluminum alloy T-joint during the dual laser beam bilateral synchronous welding. *J Manuf Process* 60:200–212
41. Liu T, Gao ZAN, Ling WL, Wang YN, Wang XM, Zhan XH (2021) Effect of heat accumulation on the microstructure of Invar alloy manufactured by multi-layer multi-pass laser melting deposition. *Opt Laser Technol* 144:107407
42. Pariona MM, Taques AF, Woiciechowski LA (2018) The Marangoni effect on microstructure properties and morphology of laser-treated Al-Fe alloy with single track by FEM: varying the laser beam velocity. *Int J Heat Mass Tran* 119:10–19
43. Chen LY, Zhao Y, Song BX, Yu TB, Liu Z (2021) Modeling and simulation of 3D geometry prediction and dynamic solidification behavior of Fe-based coatings by laser cladding. *Opt Laser Technol* 139:107009
44. Vundru C, Singh R, Yan WY, Karagadde S (2021) A comprehensive analytical-computational model of laser directed energy deposition to predict deposition geometry and integrity for sustainable repair. *Int J Mech Sci* 211:106790

45. Song BX, Yu TB, Jiang XY, Xi WC (2019) Numerical model of transient convection pattern and forming mechanism of molten pool in laser cladding. *Numer Heat Tr a-Appl* 75:855–873
46. Song BX, Yu TB, Jiang XY, Xi WC, Lin XL (2020) Effect of laser power on molten pool evolution and convection. *Numer Heat Tr a-Appl* 78:48–59
47. Wu JZ, Ren S, Zhang Y, Cao Y, Zhang DB, Yin CH (2021) Influence of spatial laser beam profiles on thermal-fluid transport during laser-based directed energy deposition. *Virtual Phys Prototy* 16:444–459
48. Dai D, Gu D (2016) Influence of thermodynamics within molten pool on migration and distribution state of reinforcement during selective laser melting of AlN/AlSi10Mg composites. *Int J Mach Tool Manu* 100:14–24
49. Wang L, Yao JH, Hu Y, Zhang QL, Sun Z, Liu R (2017) Influence of electric-magnetic compound field on the WC particles distribution in laser melt injection. *Surf Coat Tech* 315:32–43
50. Huo K, Zhou JZ, Dai FZ, Xu JL (2021) Particle distribution and microstructure of IN718/WC composite coating fabricated by electromagnetic compound field-assisted laser cladding. *Appl Surf Sci* 545:149078
51. DebRoy T, Wei HL, Zuback JS, Mukherjee T, Elmer JW, Milewski JO, Beese AM, Wilson-Heid A, De A, Zhang W (2018) Additive manufacturing of metallic components - process, structure and properties. *Prog Mater Sci* 92:112–224
52. Zhao YF, Koizumi Y, Aoyagi K, Wei DX, Yamanaka K, Chiba A (2019) Molten pool behavior and effect of fluid flow on solidification conditions in selective electron beam melting (SEBM) of a biomedical Co-Cr-Mo alloy. *Addit Manuf* 26:202–214
53. Yuan WH, Chen H, Cheng T, Wei QS (2020) Effects of laser scanning speeds on different states of the molten pool during selective laser melting: simulation and experiment. *Mater Design* 189:108542

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.